



# DataQuest Recommender System

CUSTOMER RECOMMENDATIONS

LESLEY NGCOBO

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# DATAQUEST RECOMMENDER SYSTEM ANALYSIS

## INTRODUCTION

Before building my first recommender system, I began with a careful examination of the raw dataset and identified several data quality issues that could compromise the reliability of the model. To ensure clean and meaningful inputs, I applied the following preprocessing steps:

- 1. Removed rows with missing **CustomerID** values to maintain customer integrity.
- 2. Excluded returns and cancellations, as they do not contribute to meaningful purchase patterns.
- 3. Filtered out erroneous pricing entries (where **UnitPrice**  $\leq 0$ ) to prevent distortions in item popularity calculations.

Additionally, I converted **InvoiceDate** to a standardized format (**ymd\_hms**) to support time-based analysis later in the pipeline.

With this cleaned and structured dataset, I set out to build my first recommender system, starting with a simple, global popularity-based approach that ranks items based on how often they're purchased.

## GLOBAL POPULARITY-BASED RECOMMENDER-SYSTEM

As a starting point, I developed a basic recommender system that identifies the most frequently purchased items across all users. This type of system serves as a solid baseline, especially useful in cold-start situations where user behavior data is limited. It's fast to compute, easy to interpret, and often highlights products with broad appeal.

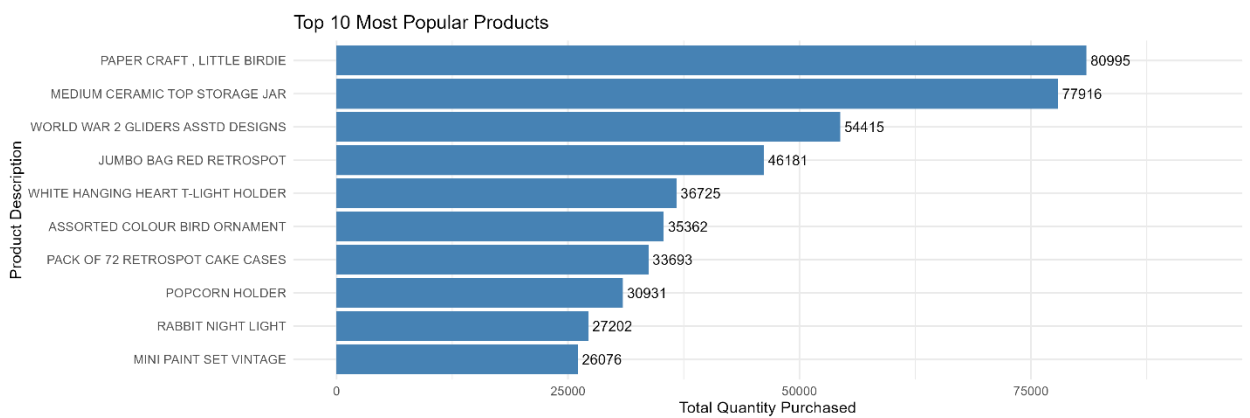


FIGURE 1: MOST POPULAR ITEMS

However, during this stage I noticed a key limitation: the vast majority of transactions in the dataset originated from the United Kingdom. This regional imbalance skewed the global popularity rankings, leading to recommendations that were heavily biased toward UK-based buying patterns.

To address this issue, I extended the model into a **group-specific popularity recommender**, first by considering geographic context (country) and later by incorporating seasonal variation (month). These adjustments aimed to enhance personalization and relevance for users in different regions and during different times of the year.

## COUNTRY-BASED POPULARITY RECOMMENDER

To better understand the geographic influence on item popularity, I visualized how the top 10 globally popular products performed across different countries.

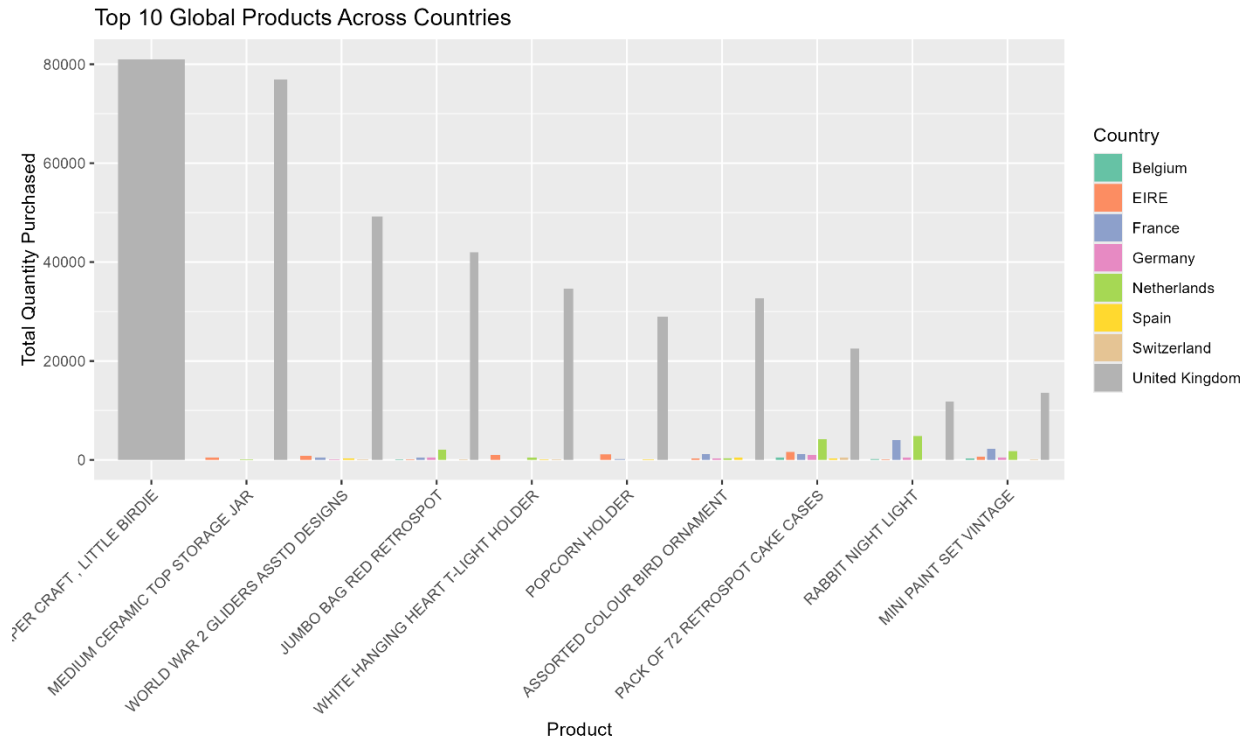


FIGURE 2.1: POPULAR PRODUCTS PER COUNTRY

As seen in the chart above, the UK dominates overall purchase volume, heavily influencing global trends. While this accurately reflects the dataset's distribution, it poses a risk for recommendation quality: if the system blindly follows global popularity, users in smaller countries may receive less relevant suggestions.

Notably, some products that appear only moderately popular globally show strong signals in specific regions. For instance, items like **RABBIT NIGHT LIGHT** or **MINI PAINT SET VINTAGE** emerge as local favorites in countries such as Germany and the Netherlands. These long-tail regional trends might be lost in a global model but are surfaced through country-specific analysis.

To capitalize on this, I implemented a **country-level popularity recommender**, which tailors recommendations to a user's country by surfacing the top items most frequently purchased in that region. This allows the system to reflect local preferences more effectively and boosts its potential to deliver relevant and personalized product suggestions.

## COUNTRY + MONTH POPULARITY RECOMMENDER

While analyzing country-based trends, I started to consider another familiar pattern: some products are more popular at specific times of the year. Just as I personally tend to buy festive items in December or home decor during seasonal transitions, I wanted to test whether similar seasonal trends were evident in this dataset.

To explore this, I examined the *monthly purchase patterns of the top five globally popular products* across the five most active countries.

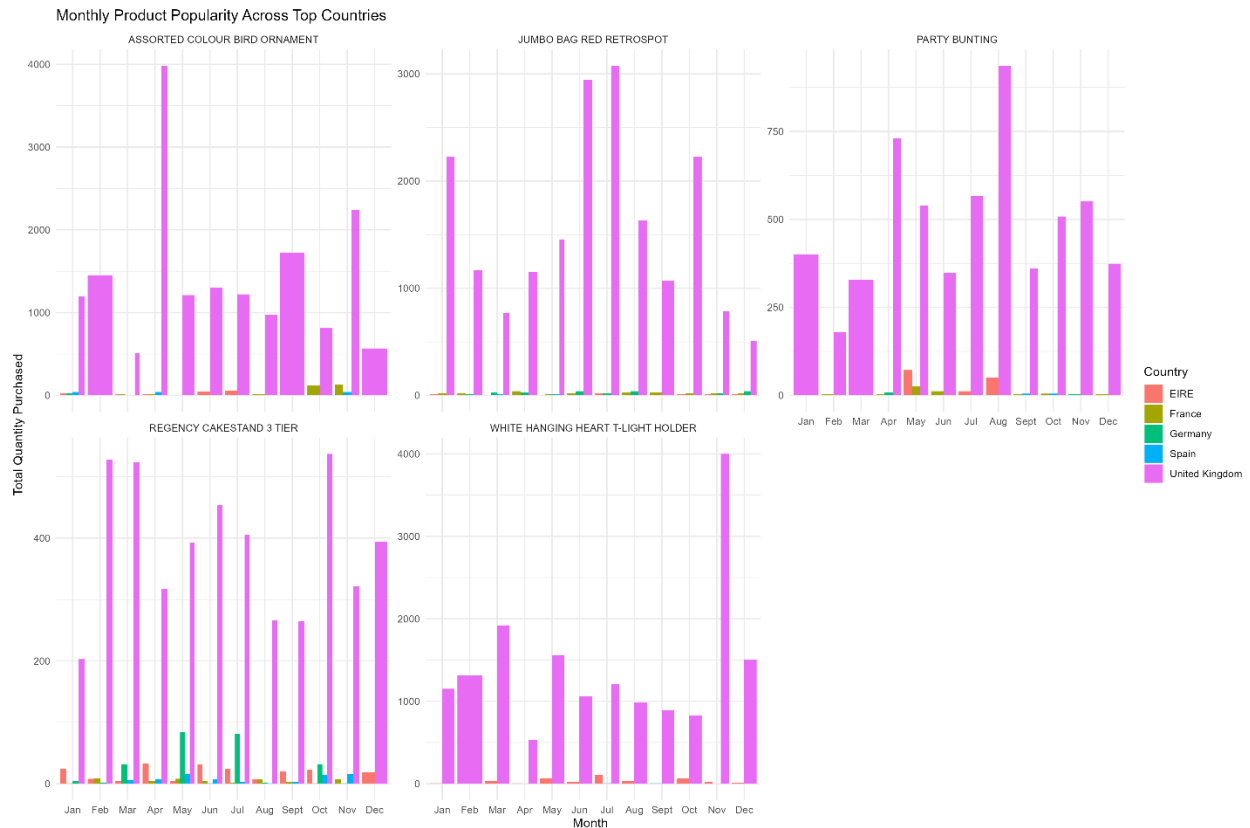


FIGURE 2.2: SEASONAL POPULARITY

The results confirmed my assumption, none of the top products are purchased at a consistent rate throughout the year. For example, several items spike in **November and December**, likely reflecting holiday shopping behavior, while others show surges mid-year or drop off sharply in particular months. This strongly suggests that *seasonality influences customer behavior*.

Relying only on global or even country-level popularity would overlook these temporal nuances. Recommendations based on such averages could miss seasonal purchasing intent, especially around holidays or specific events.

To account for this, I designed a *Month-Aware Popularity Recommender*, which suggests products based on *what's trending in a customer's country during the current month*. This temporal layer brings the model closer to real-world buying behavior—offering more relevant, time-sensitive product suggestions.

One standout example is the **WHITE HANGING HEART T-LIGHT HOLDER**, which shows a sharp rise in purchases during **November and December**, particularly in the **UK and Germany**, likely linked to winter decor or festive gifting. Similarly, the **RABBIT NIGHT LIGHT** shows higher activity in **March and April**, which may align with **springtime events like Easter**.

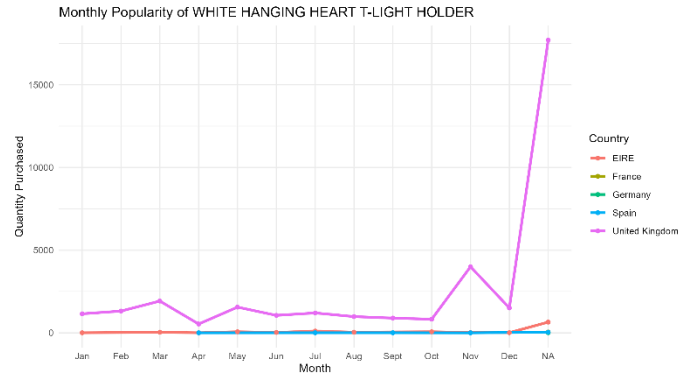


FIGURE 2.3

By incorporating these seasonal signals, this recommender system adapts more dynamically to changing customer interests, enabling **smarter, context-aware recommendations** that feel timely and relevant.

## COMPARISON OF THE THREE POPULARITY-BASED RECOMMENDER SYSTEMS

As we progress through the popularity-based models, a clear pattern emerges; **recommendation performance improves as we incorporate more context**. Starting with a basic global model, then adding **geographic specificity**, and finally **seasonal awareness**, each iteration makes the system more relevant to the customer's environment.

**PERFORMANCE COMPARISON TABLE**

| Recommender Type         | Context Considered              | Example Logic                                           | Precision@5 (Sample/Estimated) | Recall@5 (Sample/Estimated) |
|--------------------------|---------------------------------|---------------------------------------------------------|--------------------------------|-----------------------------|
| Global Popularity        | None                            | Top products overall                                    | 0.0021                         | 0.0106                      |
| Country-Based Popularity | Geographic (customer's country) | Top products in customer's country                      | 5e-04                          | 0.0024                      |
| Month + Country-Based    | Geographic + Temporal           | Top products in customer's country during current month | 0.0035                         | 0.0177                      |

TABLE 1: PERFORMANCE COMPARISON OF THE 3 POPULARITY SYSTEMS

**Precision@5** measures how many of the top 5 recommended items were actually relevant to the **user**. **Recall@5** measures how many of the items the user actually bought were captured in the top 5 recommendations.

## GLOBAL POPULARITY RECOMMENDER

The global popularity model, which recommends the five most purchased items overall, **serves as a simple baseline**. While broadly appealing, its lack of personalization limits its effectiveness.

- **Precision@5: 0.21%** – Only a tiny fraction of recommendations matched what the customer actually purchased.
- **Recall@5: 1.06%** – Most of the customer's real purchases were not captured by the top 5 suggestions.

This model reflects the risks of relying on global averages in datasets with heavy **regional or seasonal bias**. It sets a useful baseline for measuring improvement in more personalized systems.

## COUNTRY-BASED POPULARITY RECOMMENDER

This model introduces **regional awareness**, recommending the most popular products in a customer's country. Despite expectations, it **slightly underperforms** the global model:

- **Precision@5: 0.05%**
- **Recall@5: 0.24%**

This underperformance may stem from **smaller countries** having too few users or product diversity, making their top items less relevant overall. However, in **larger regions like the UK**, country-specific recommendations still outperform global ones qualitatively. The model succeeds at **regional personalization**, even if that doesn't always translate into higher metrics.

## COUNTRY + MONTH POPULARITY RECOMMENDER

This third system introduces a **temporal layer**, accounting for **seasonal shifts** in purchase behavior. It recommends items that are both popular in the customer's **country and current month**.

- **Precision@5: 0.35%**
- **Recall@5: 1.77%**

This performance jump highlights the **value of time-based context**; customers are more likely to buy products that are locally and seasonally trending. For example, **festive decor in December** or **spring-themed items in April**. Adding this temporal signal boosts the system's ability to recommend products that match **real-world buying behavior**.

## TAKEAWAY: CONTEXT MATTERS

This comparison demonstrates a simple but powerful principle:

**The more relevant context a recommender understands, the better it performs.**

- The **Global model** offers stability but no personalization.
- The **Country model** improves regional relevance but struggles in smaller markets.
- The **Country + Month model** incorporates both space and time, offering the **most personalized and dynamic popularity-based recommendations** in this set.

This progression also sets the stage for **even more personalized approaches** like **Collaborative Filtering**, which learns directly from individual customer behavior, and **Hybrid models**, which combine multiple signals for deeper relevance.

## COLLABORATIVE FILTERING

Until this point, my recommendation systems have been general-purpose, they relied on aggregate popularity to suggest products to users. But customer preferences are highly individual, and what's popular overall may not be relevant to a specific person.

To address this, I turned to **Collaborative Filtering**, a more personalized approach that recommends products based on the behavior of similar users or the relationships between items. Rather than relying on what's popular, collaborative filtering infers recommendations from **patterns in user behavior**.

### USER-BASED COLLABORATIVE FILTERING (UBCF)

UBCF recommends items based on what similar users have purchased, following a **“people like you also bought...”** logic. To implement this, I computed **cosine similarity scores** between users, identifying those whose purchase histories overlap significantly.

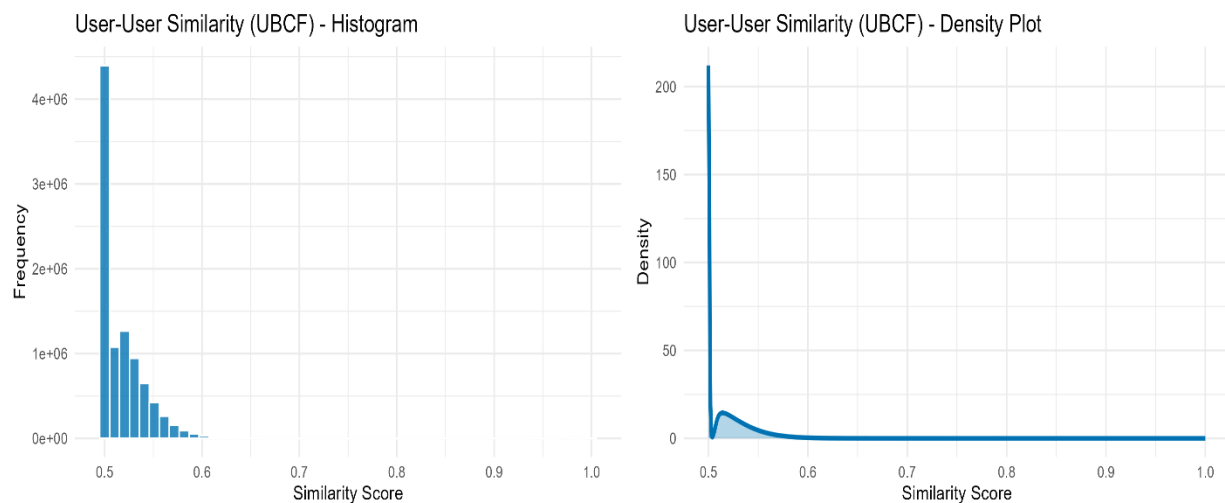


FIGURE 3.1: DISTRIBUTION OF USER SIMILARITIES

The chart shows a **left-skewed distribution**, with a strong peak in the **0.45–0.55** range. This suggests that while some users do share moderate similarities, most have **very low overlap** in their purchasing behavior.

Notably:

- Similarity scores above **0.6** are rare.
- There is **near-zero density** beyond **0.65**, meaning strong user-to-user similarity is uncommon.
- Yet these rare high-similarity users are **critical**, as UBCF relies on them (nearest neighbors) to generate personalized recommendations.



This **sparsity** presents a challenge: many users have unique or fragmented purchase histories, which limits UBCF's effectiveness in highly diverse datasets. Nevertheless, when strong user similarities are present, the model can make highly targeted and relevant suggestions.

## ITEM-BASED COLLABORATIVE FILTERING (IBCF)

IBCF works from a different angle, it doesn't compare users, but rather items. The core logic is **“people who bought this also bought...”**, based on how often products are purchased together across many users.

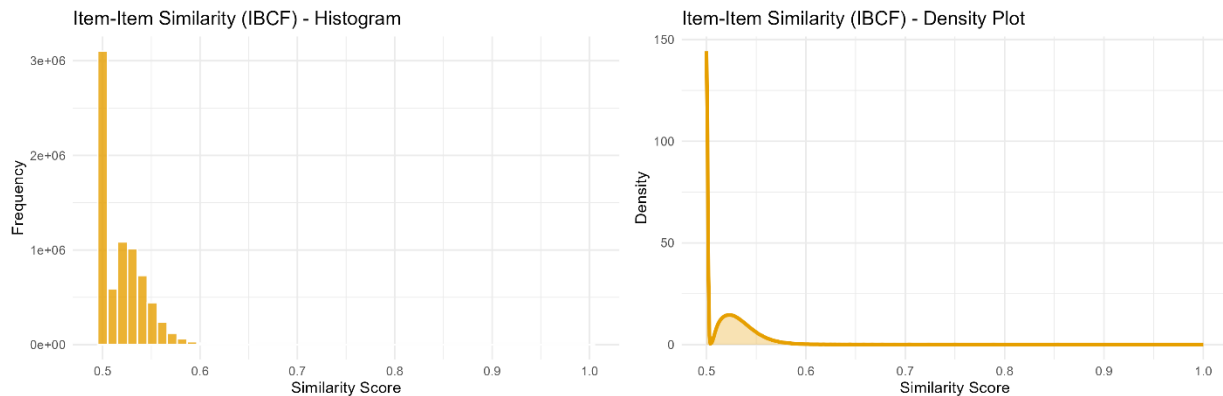


FIGURE 3.2: DISTRIBUTION OF ITEM SIMILARITY

Like UBCF, the item similarity distribution is also **left-skewed**, but it features a **distinct peak** around **0.51–0.54**. This peak reveals a meaningful cluster of **item pairs that tend to be co-purchased**, for example, items that are part of a set, complementary decor, or bundled gifts.

Key insights:

- While high item similarity (>0.6) is still rare, it's **more common than high user similarity**.
- These item-item relationships are often **more stable** than user preferences.
- IBCF performs better in **sparse user environments**, as it leverages shared purchase patterns across many users, even if those users have different profiles.

In short, . If many people buy item A and item B together, IBCF can confidently recommend B to anyone who buys A, regardless of who they are.

## COMPARISON OF COLLABORATIVE FILTERING MODELS VS POPULARITY-BASED APPROACHES

Collaborative Filtering (CF) methods personalize recommendations by analyzing past user behavior. These models infer preferences by comparing either:

- **Users to other users** (User-Based CF), or
- **Items to other items** (Item-Based CF).

The assumption is that this behavioral personalization should outperform broader, group-based methods like popularity recommenders. But how does this hold up in practice?

The table below compares each approach using **Precision@10** and **Recall@10**, which measure how many recommended items were relevant and how much of a user's actual behavior was captured, respectively.

**PERFORMANCE COMPARISON TABLE**

| Recommender Type                          | Context Considered    | Example Logic                                           | Precision@10 (Sample/Estimated) | Recall@10 (Sample/Estimated) |
|-------------------------------------------|-----------------------|---------------------------------------------------------|---------------------------------|------------------------------|
| <i>Month + Country-Based</i>              | Geographic + Temporal | Top products in customer's country during current month | 0.0021                          | 0.021                        |
| <i>User-Based Collaborative Filtering</i> | User-User Similarity  | "People like you also bought..."                        | 0.0014                          | 0.0141                       |
| <i>Item-Based Collaborative Filtering</i> | Item-Item Similarity  | "People who bought this also bought..."                 | 0.0174                          | 0.1741                       |

**TABLE 2: PERFORMANCE COMPARISON OF THE COLABORATIVE FILTERING MODELS AND HYBRID-POPULARITY SYSTEMS**

## INSIGHTS

**User-Based CF (UBCF)** performs the weakest of the three, with a **Precision@10 of 0.14%** and **Recall@10 of 1.41%**. While UBCF introduces personalization, it suffers from **sparse user overlap** because most customers don't buy the same products. Without strong user similarity, it becomes difficult for the model to generate accurate suggestions, despite its personalized approach.

**Item-Based CF (IBCF)** outperforms both UBCF and popularity-based models. With a **Precision@10 of 1.74%** and **Recall@10 of 17.41%**, it effectively leverages **item co-purchase patterns** to make reliable suggestions. Even when users differ, **products that are frequently bought together** can be confidently recommended, making IBCF more scalable and effective.

**Country + Month Popularity** slightly edges out UBCF but still trails behind IBCF. With a **Precision@10 of 0.21%** and **Recall@10 of 2.10%**, it captures **regional and seasonal relevance** but lacks **personalization**, recommending the same items to all users in a group.

## KEY TAKEAWAY

These results show a clear pattern: the more behavior-specific context a model captures, the better its performance. While **popularity-based systems** are simple and interpretable, their generality limits precision. **Collaborative filtering**, especially **IBCF**, benefits from rich behavioral signals and offers significantly improved personalization.

However, each model still has gaps:

- UBCF struggles with sparse user overlap.
- IBCF doesn't consider individual preferences or context.
- Popularity systems ignore behavior entirely.

This sets the stage for **hybrid recommender systems**, which aim to combine the strengths of all three personalization, co-purchase logic, and contextual relevance to deliver **richer, more adaptive recommendations**.

## WEIGHTED HYBRID RECOMMENDER

The **Weighted Hybrid Recommender** combines the strengths of two collaborative filtering approaches:

- **User-Based Collaborative Filtering (UBCF)**, which captures personalized behavior by finding customers with similar purchasing patterns.
- **Item-Based Collaborative Filtering (IBCF)**, which leverages co-purchase patterns between products.

These two models each generate their own top-N recommendation lists. The hybrid model then merges them using a **weighted voting mechanism** in this case, **70% IBCF and 30% UBCF**.

*My initial hybrid combined UBCF and IBCF with equal weight (i.e., simple score averaging), but performance was suboptimal.*

*This led to the development of the weighted hybrid, where IBCF's stronger precision and recall earned a higher vote share, resulting in improved relevance and recommendation quality.*

In this setup, each model's recommendations act as "votes":

- Items suggested by IBCF are amplified with more weight (70%)
- UBCF recommendations contribute 30% of the influence.

This balance enables the hybrid system to maintain **precision and robustness from IBCF**, while incorporating the **personalized nuance of UBCF**, resulting in a more well-rounded recommender.

## HOW THE MODEL WORKS

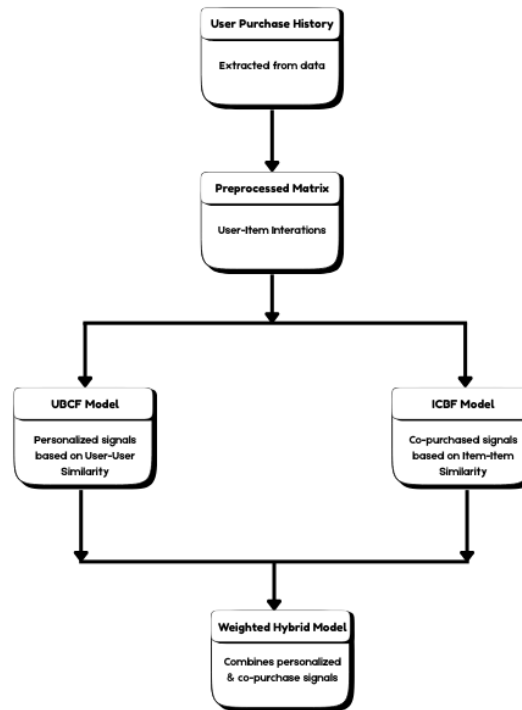


FIGURE 4.1: STRUCTURE OF THE HYBRID MODEL

The flowchart above illustrates the hybrid model architecture:

1. **User Purchase History:** Raw transaction data captures each customer’s past purchases, serving as the behavioral foundation.
2. **User-Item Interaction Matrix:** Transactions are transformed into a binary matrix showing which items each user has interacted with. This feeds both UBCF and IBCF models.
3. **UBCF Model – Personalized Signals:** Identifies similar users based on overlapping purchase histories and recommends items those users bought but the target user hasn’t.
4. **IBCF Model – Co-Purchase Signals:** Identifies strong item-to-item relationships, suggesting products commonly purchased together, regardless of user similarity.
5. **Weighted Merging of Recommendations:** Each model outputs a top-N recommendation list. The hybrid combines these lists using a **weighted voting mechanism**, with:

- a. 70% weight for IBCF
- b. 30% weight for UBCF

This means that a recommendation supported by IBCF contributes more to the final ranking than one supported by UBCF. These weights weren’t arbitrary, they reflect actual performance metrics, where IBCF consistently outperformed UBCF in both precision and recall. The 70/30 split balances **stability and relevance** (from IBCF) with **personalized diversity** (from UBCF).

## PERFORMANCE COMPARISON: WEIGHTED VS. BASELINE HYBRID

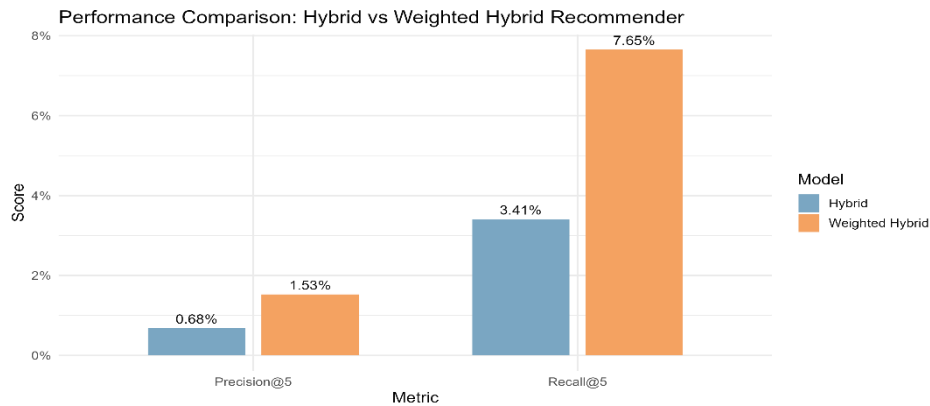


FIGURE 4.2: COMPARISON OF NON-WEIGHTED AND WEIGHTED HYBRID MODELS

The weighted model delivers a noticeable performance boost. By giving more influence to IBCF, without discarding UBCF's personalization, it creates *more relevant and well-rounded recommendations*. This result also supports the value of *fine-tuning hybrid models*, rather than simply merging models equally.

## FULL WEIGHTED RECOMMENDER SYSTEM

The **Full Weighted Hybrid Recommender** represents the most comprehensive and context-aware model in this project. It integrates four distinct recommendation signals, each capturing a different dimension of customer behavior to deliver more accurate, dynamic, and practical recommendations.

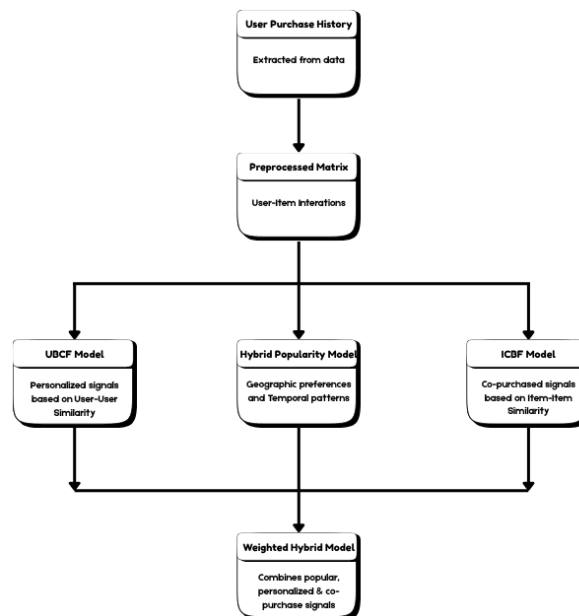


FIGURE 5.1: FULL WEIGHTED HYBRID SYSTEM STRUCTURE

WHAT THE SYSTEM COMBINES

As shown in Figure X, this model draws from:

- **User-Based Collaborative Filtering (UBCF):** Captures personal preferences by identifying users with similar purchasing behavior.  
“What have people like you purchased?”
- **Item-Based Collaborative Filtering (IBCF):** Recommends products based on co-purchase patterns across the dataset.  
“People who bought this also bought...”
- **Country-Based Popularity:** Adds geographic relevance by recommending the most popular items in the user’s region.
- **Month-Based Popularity:** Infuses temporal awareness by adapting recommendations to seasonal trends and holiday cycles.

Each model contributes a different type of “signal”—from personal behavior to broader seasonal and regional trends.

HOW THE MODELS WORK

Each model produces a ranked list of product recommendations. These lists are then merged using a **weighted voting system**, where some models have more influence than others based on their standalone performance:

| Component              | Weight (%) |
|------------------------|------------|
| IBCF (Item Similarity) | 50%        |
| UBCF (User Similarity) | 20%        |
| Country Popularity     | 20%        |
| Month Popularity       | 10%        |

TABLE 3: WEIGHT VOTING MECHANISM

These weights weren’t randomly assigned—they were informed by the performance of each model during evaluation. **IBCF**, having consistently delivered the strongest results, holds the majority vote. At the same time, UBCF and popularity-based signals are included to increase personalization, diversity, and context-awareness.

*This isn't just a combination of models, it's a strategic balance of signals. If collaborative filters underperform due to data sparsity, the model can lean on popularity trends to maintain relevance.*

WHY USE THIS HYBRID?

This model performs slightly below IBCF in raw metrics (Precision@5: **1.95%**, Recall@5: **9.76%** vs. **IBCF’s 2.2% and 10.9%**). However, it **outshines all others in practical value**:

- Personalized (via UBCF)
- Co-purchase smart (via IBCF)
- Regionally adaptive (via Country Popularity)

- Seasonally aware (via Month Popularity)

Instead of optimizing for only accuracy, this model optimizes for **real-world relevance**, it understands not just *what* customers tend to buy, but *when*, *where*, and *why* they do so.

*This is the system I would trust most in a live production setting.*

## PERFORMANCE COMPARISON

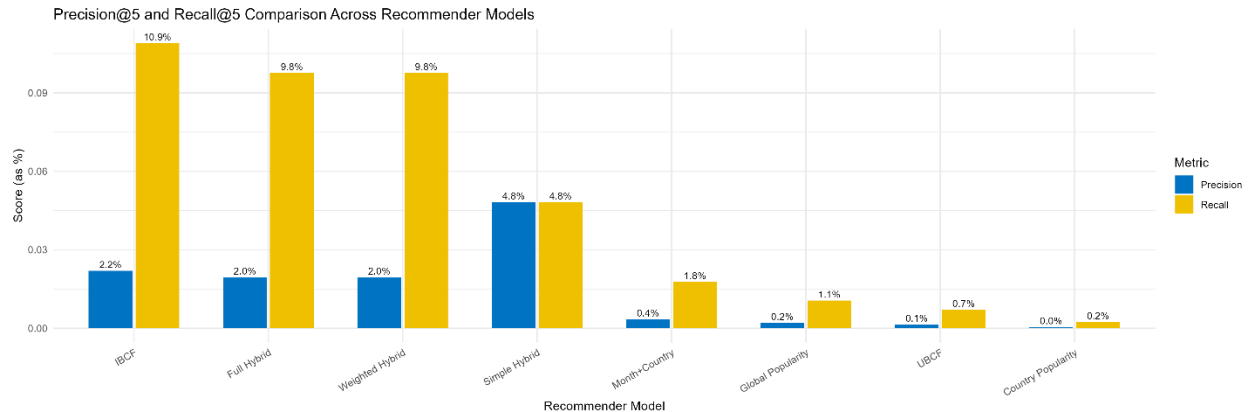


FIGURE 5.2: COMPARISON OF ALL THE MODELS

In **Figure 5.2**, the full hybrid model demonstrates strong performance, trailing IBCF slightly in pure recall and precision but offering **greater versatility and robustness**. Unlike IBCF, which lacks temporal or geographic nuance, this model responds to seasonal shifts, regional preferences, and individual behavior patterns—all in one.

## CONCLUSION

This project set out to build a product recommendation system by starting with simple models and gradually layering in complexity. From global popularity to user- and item-based collaborative filtering, and finally to hybrid systems, each step brought new insights and capabilities.

Throughout this process, it became clear that **no single method is universally best**. Popularity models are fast but generic, while collaborative filtering adds personalization but suffers from sparse data. The **Full Weighted Hybrid Recommender** emerged as the most balanced solution, blending user behavior, item relationships, regional trends, and seasonal patterns into one adaptable system.

Although it didn't achieve the highest precision or recall, it proved to be the most context-aware and scalable model. It reflects real-world buying behavior more holistically and offers a practical path toward dynamic, personalized recommendations.

Beyond performance, this project taught me how to weigh trade-offs, design adaptable systems, and focus on recommendations that are not just accurate, but relevant, interpretable, and usable in real business settings.